

The Set-Theoretic Perspective: Numbers as von Neumann Ordinals

Paper 02 of the Univalent Correspondence Series

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Abstract

We examine the second of six perspectives in the Univalent Correspondence series: the set-theoretic level, in which natural numbers are reduced to particular elements of the cumulative hierarchy of pure sets. We work primarily in Zermelo–Fraenkel set theory with Choice (ZFC), focusing on John von Neumann’s ordinal encoding $0 = \emptyset$, $n + 1 = n \cup \{n\}$, where the natural number 58 is realized as the set $\{0, 1, \dots, 57\}$ of all smaller von Neumann ordinals. We contrast this with Ernst Zermelo’s earlier encoding $0 = \emptyset$, $n + 1 = \{n\}$, in which 58 is realized as a chain of 58 nested singletons. Both encodings produce structures isomorphic to $(\mathbb{N}, 0, \text{succ})$ as Peano models, yet they differ on *internal* set-theoretic questions such as whether $2 \in 3$ or whether $\{\emptyset\} \in 17$. Following Paul Benacerraf’s 1965 paper *What Numbers Could Not Be*, we argue that this discrepancy is not a defect to be eliminated but a structural fact: numbers are not sets, and any attempt to identify them with sets generates encoding-relative *junk theorems* that are meaningful about the encoding but meaningless about the number. We give a careful presentation of the ZFC axiomatization, the cumulative hierarchy V_α , ranks, and well-foundedness; we define addition, multiplication, and order recursively on the von Neumann ordinals and verify that these definitions satisfy the Peano axioms; and we prove a general representation theorem stating that any successor structure obtained from a class function S that is injective, has a base point not in its image, and admits transfinite iteration generates a model of Peano arithmetic isomorphic to the von Neumann naturals. We close with a structuralist re-reading: the set-theoretic perspective *does* present numbers, but only as encodings; the structural content these encodings share is a universal property, which Paper 03 will make explicit. Our development is accompanied by a Haskell implementation that realizes both encodings as distinct types and proves, by construction, that they satisfy the same Peano interface.

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1 Introduction

1.1 Where we are in the series

This is the second paper in a six-paper series, plus a synthesis volume, exploring the question “*what is a natural number?*” at progressively higher levels of mathematical abstraction. Paper 01 considered the *naïve* level, in which a number is treated as a primitive symbol or quantity, without further analysis. The naive level is the practical starting point of arithmetic and the inevitable surface of any numeral system, but it conflates syntax with semantics: the symbols “58”, “LVIII”, and “ $\underbrace{|\cdot\cdot\cdot|}_{58}$ ” all refer to the same number, but the naive level

provides no machinery for distinguishing the numeral from the number it names.

The natural next step, taken in the foundational programs of the early twentieth century, is to *reduce* numbers to objects of a more basic theory. Frege [3] attempted such a reduction to logic; Russell and Whitehead refined and elaborated it in *Principia Mathematica* [10]; Zermelo and later Fraenkel and Skolem developed the axiomatic set theory we now call ZFC. Within ZFC, John von Neumann [14] gave a canonical recursive definition of the ordinals that, when restricted to the finite case, produces a particular set-theoretic implementation of the natural numbers. Standard expositions of the resulting machinery include Halmos [4], Suppes [13], Shoenfield [11], and Kunen [5].

1.2 The thesis of this paper

We make four claims:

1. **Set theory does provide a foundation for arithmetic.** The von Neumann encoding inside ZFC produces a structure $(\omega, 0, \text{succ})$, where ω is the smallest inductive set, on which addition, multiplication, and order can be defined recursively, and which satisfies the (second-order) Peano axioms.
2. **The encoding is not unique.** Zermelo’s earlier encoding, $0 = \emptyset$, $n + 1 = \{n\}$, gives a different set-theoretic structure that nevertheless also satisfies the Peano axioms. Many other encodings exist. The encodings are mutually *Peano-isomorphic* but not mutually *ZFC-isomorphic*: as sets, they differ.
3. **The discrepancy generates junk theorems.** In von Neumann’s encoding, $2 \in 3$ is true; in Zermelo’s, it is false. The question “is 2 an element of 3?” is meaningful inside ZFC for each encoding separately, but it is meaningless about *the number 2* — it is an artifact of the encoding. Following Benacerraf [1], we conclude that numbers cannot be identified with sets.
4. **The structural content survives.** What both encodings share is precisely the universal property of a Natural Number Object: they are both initial in a category of pointed sets with endomorphism. This is the structural content that Paper 03 will make explicit. The set-theoretic perspective is therefore not refuted but *relocated*: it is one implementation of a structural specification, not a definition.

1.3 Outline

Section 2 reviews ZFC, with particular attention to the axiom of infinity, the axiom schema of replacement, and the foundation axiom (which underwrites well-foundedness). Section 3 presents the von Neumann encoding, defines ω , and verifies the Peano axioms. Section 4 presents Zermelo’s encoding and contrasts the two. Section 5 develops Benacerraf’s identification problem and the notion of a junk theorem. Section 6 defines addition, multiplication, and order recursively and proves the basic algebraic identities. Section 7 introduces the cumulative hierarchy V_α , ranks, and well-foundedness, and locates the natural numbers within it. Section 8 proves a representation theorem characterizing successor structures up to isomorphism. Section 9 develops the structuralist re-reading and sketches the bridge to Paper 03. The Haskell implementation accompanying the paper realizes both encodings; Section 10 describes its design and the Peano-axiom verification by construction.

2 Mathematical Framework: ZFC Reviewed

We work in classical first-order logic with equality and a single binary relation symbol $\in$.

2.1 Axioms

Definition 2.1 (ZFC). The axioms of Zermelo–Fraenkel set theory with Choice are:

(Ext) *Extensionality*. $\forall x \forall y (\forall z (z \in x \leftrightarrow z \in y) \rightarrow x = y)$.

(Pair) *Pairing*. $\forall x \forall y \exists z \forall w (w \in z \leftrightarrow w = x \vee w = y)$.

(Un) *Union*. $\forall x \exists y \forall z (z \in y \leftrightarrow \exists w (w \in x \wedge z \in w))$.

(Pow) *Power set*. $\forall x \exists y \forall z (z \in y \leftrightarrow z \subseteq x)$.

(Sep) *Separation schema*. For each formula $\varphi(z)$ not involving y , $\forall x \exists y \forall z (z \in y \leftrightarrow z \in x \wedge \varphi(z))$.

(Repl) *Replacement schema*. For each formula $\varphi(x, y)$ defining a class function F ,

$$\forall A \exists B \forall y (y \in B \leftrightarrow \exists x \in A \varphi(x, y)).$$

(Inf) *Infinity*. $\exists I (\emptyset \in I \wedge \forall x (x \in I \rightarrow x \cup \{x\} \in I))$.

(Found) *Foundation*. $\forall x (x \neq \emptyset \rightarrow \exists y \in x (y \cap x = \emptyset))$.

(AC) *Choice*. Every set of nonempty sets has a choice function.

Remark 2.2. The empty set $\emptyset$ is definable: by Separation applied to any set x and the formula $z \neq z$, the resulting set has no elements. By Extensionality, this set is unique. **(Inf)** guarantees there is a set containing $\emptyset$ and closed under $x \mapsto x \cup \{x\}$.

2.2 Axiom of Infinity, in detail

Definition 2.3 (Inductive set). A set I is *inductive* when $\emptyset \in I$ and, for every $x \in I$, $x \cup \{x\} \in I$. The axiom of infinity asserts that an inductive set exists.

Proposition 2.4 (Existence of ω). *There exists a unique smallest inductive set, denoted ω . Equivalently, $\omega = \bigcap \{I : I \text{ is inductive}\}$ inside any inductive set, defined by Separation:*

$$\omega = \{x \in I_0 : \forall I (I \text{ inductive} \rightarrow x \in I)\}, \quad (1)$$

where I_0 is any fixed inductive set guaranteed by **(Inf)**.

Proof. The class of inductive sets is nonempty by **(Inf)**; the class-intersection is inductive (it contains $\emptyset$ since each inductive set does, and is closed under $x \mapsto x \cup \{x\}$ since each inductive set is). Separation applied to I_0 and the formula $\varphi(x) \equiv \forall I (\text{Ind}(I) \rightarrow x \in I)$ produces a set ω with the required property. Uniqueness follows from Extensionality. $\square$

2.3 Replacement, and what it buys us

The Separation schema only carves out subsets of an existing set; it cannot generate sets “larger” than what is already given. Replacement is strictly stronger: it asserts that the image of a set under a definable class function is a set. This is essential for transfinite recursion (and hence for the cumulative hierarchy of Section 7) and for the formal definition of operations such as addition by recursion on ω .

Remark 2.5. ZFC without Replacement is Zermelo’s original 1908 system Z, which is too weak for many constructions used in ordinary mathematics (Borel hierarchies, transfinite recursion past $\omega + \omega$, etc.). For our purposes here — defining the natural numbers and their arithmetic — Replacement is the crucial axiom that lets us define functions by recursion.

2.4 Foundation and well-foundedness

Definition 2.6 (Well-founded relation). A relation R on a class C is *well-founded* when every nonempty subset $S \subseteq C$ has an R -minimal element: an $x \in S$ with $\{y \in S : yRx\} = \emptyset$.

Theorem 2.7. *The Foundation axiom is equivalent (in ZF without Foundation) to the assertion that the membership relation $\in$ is well-founded on the universe of sets. In particular, there is no infinite descending $\in$ -chain $\cdots \in x_2 \in x_1 \in x_0$, no set x with $x \in x$, and no cycle $x_0 \in x_1 \in \cdots \in x_n \in x_0$.*

Proof sketch. If $x \in x$, then $\{x\}$ has no $\in$ -minimal element, contradicting Foundation. The general statement is proved by considering the transitive closure of any candidate descending chain and applying Foundation to the resulting set. $\square$

Foundation is what allows transfinite induction on the entire set-theoretic universe. It is also what makes the von Neumann encoding (in which n is literally the set of its predecessors) compatible with the standard machinery: since each n is a transitive set linearly ordered by $\in$, Foundation guarantees we never run into pathologies like $n \in n$.

3 The von Neumann Encoding

3.1 Definition

Definition 3.1 (Successor). For any set x , the *successor* of x is $x^+ := x \cup \{x\}$.

Definition 3.2 (Von Neumann naturals). We define the von Neumann naturals as elements of ω recursively:

$$0_{\text{vN}} := \emptyset, \tag{2}$$

$$(n+1)_{\text{vN}} := (n_{\text{vN}})^+ = n_{\text{vN}} \cup \{n_{\text{vN}}\}. \tag{3}$$

Unrolling this definition:

$$\begin{aligned} 0 &= \emptyset \\ 1 &= \{0\} = \{\emptyset\} \\ 2 &= \{0, 1\} = \{\emptyset, \{\emptyset\}\} \\ 3 &= \{0, 1, 2\} = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} \\ &\vdots \\ n &= \{0, 1, \dots, n-1\} \end{aligned}$$

Example 3.3. The number 58, in von Neumann form, is the set of all smaller naturals:

$$58_{\text{vN}} = \{0, 1, 2, \dots, 57\}. \tag{4}$$

The set 58 has exactly 58 elements, and each of those elements is a set with strictly fewer elements. The total cumulative count of $\in$ -occurrences inside 58 is large but finite (combinatorially, $\sum_{k=0}^{57} k = 1653$).

3.2 Membership and subset coincide

A signal feature of the von Neumann encoding is the conflation, at the finite level, of $\in$ and $\subseteq$.

Proposition 3.4. *For all $m, n \in \omega$, we have $m \in n$ if and only if $m \subsetneq n$.*

Proof. We prove both directions by induction on n .

($\Rightarrow$). If $m \in n$, then since n is the set of its predecessors and is transitive (Theorem 3.5 below), every element of m is also an element of n , hence $m \subseteq n$. Strict inclusion follows from $m \in n$ and $m \notin m$ (Foundation, Theorem 2.7).

($\Leftarrow$). If $m \subsetneq n$, pick any $k \in n \setminus m$ of minimal $\in$ -rank. Then $k \subseteq m$ (by minimality), and $k = m$ follows from extensionality together with $m \subseteq n$ and the definition of n as the set of its predecessors. Hence $m = k \in n$. $\square$

Proposition 3.5 (Transitivity of naturals). *Each $n \in \omega$ is a transitive set: if $x \in n$, then $x \subseteq n$.*

Proof. By induction. $0 = \emptyset$ is vacuously transitive. Assume n is transitive and let $x \in n^+ = n \cup \{n\}$. If $x = n$, then $x = n \subseteq n^+$ since $n \cup \{n\} \supseteq n$. If $x \in n$, then by the inductive hypothesis $x \subseteq n \subseteq n^+$. $\square$

Example 3.6. $2 \in 3$ holds in von Neumann: $3 = \{0, 1, 2\}$, so 2 is literally one of the three elements of 3. Equivalently, $2 = \{0, 1\} \subsetneq \{0, 1, 2\} = 3$. This is the kind of statement that, by Theorem 3.4, is equivalent in von Neumann to the order $2 < 3$.

3.3 Peano axioms for the von Neumann naturals

Theorem 3.7. *The structure $(\omega, 0_{\text{vN}}, (\cdot)^+)$ satisfies the Peano axioms:*

1. $0_{\text{vN}} \in \omega$.
2. For all $n \in \omega$, $n^+ \in \omega$.
3. For all $n \in \omega$, $n^+ \neq 0_{\text{vN}}$.
4. For all $m, n \in \omega$, if $m^+ = n^+$ then $m = n$.
5. (Induction) For every set $S \subseteq \omega$, if $0_{\text{vN}} \in S$ and S is closed under $(\cdot)^+$, then $S = \omega$.

Proof. (1) and (2) are immediate from the definition of ω as the smallest inductive set.

(3) If $n^+ = \emptyset$, then $n \cup \{n\} = \emptyset$. But $n \in n \cup \{n\}$, contradiction.

(4) Suppose $m^+ = n^+$, i.e. $m \cup \{m\} = n \cup \{n\}$. Then $m \in n^+$ and $n \in m^+$. By Theorem 3.4 (applied via $\in$ -trichotomy), if $m \neq n$ then either $m \in n$ and $n \in m$, which contradicts Foundation, or one is a proper subset of the other; combined with $m \in n^+$ and $n \in m^+$ this forces $m = n$.

(5) Let $S \subseteq \omega$ be inductive (containing 0 and closed under $(\cdot)^+$). Then S itself is an inductive set, so $\omega \subseteq S$ by minimality of ω . The reverse inclusion is by hypothesis. $\square$

4 Zermelo's Encoding

Ernst Zermelo, in his 1908 axiomatization [16], used a different recursive definition of the natural numbers.

Definition 4.1 (Zermelo naturals). We define the Zermelo naturals recursively:

$$\begin{aligned} 0_Z &:= \emptyset, \\ (n+1)_Z &:= \{n_Z\}. \end{aligned} \tag{5} \tag{6}$$

Unrolling:

$$\begin{aligned}
0_Z &= \emptyset \\
1_Z &= \{\emptyset\} \\
2_Z &= \{\{\emptyset\}\} \\
3_Z &= \{\{\{\emptyset\}\}\} \\
&\vdots \\
n_Z &= \underbrace{\{\{\dots\{\emptyset\}\dots\}\}}_n \underbrace{\{\dots\}\}_n
\end{aligned}$$

Example 4.2. 58_Z is a chain of 58 nested singletons:

$$58_Z = \underbrace{\{\{\dots\{\emptyset\}\dots\}\}}_{58} \underbrace{\{\dots\}\}_{58}. \quad (7)$$

Crucially, 58_Z has exactly *one* element (namely 57_Z), not 58. The cardinality of the encoding does not match the natural number it encodes.

4.1 Comparison: junk theorems

Proposition 4.3 (Encoding-relative facts). *The following statements all have determinate truth values inside ZFC, but the answers depend on the encoding. We display together the cases where the encodings agree and where they differ:*

<i>Question</i>	<i>Von Neumann</i>	<i>Zermelo</i>
$0 \in 1?$	<i>true</i>	<i>true</i> (both: $1 = \{0\}$)
$n \in n^+?$	<i>true</i>	<i>true</i> (by construction)
$n \subseteq n^+?$	<i>true</i>	<i>true</i>
$1 \in 3?$	<i>true</i> ($3 = \{0, 1, 2\}$)	false ($3_Z = \{2_Z\}$)
$1 \in 17?$	<i>true</i>	false
$\{\emptyset\} \in 17?$	<i>true</i> (since $\{\emptyset\} = 1 \in 17$)	false
$ n = n$ for $n \in \omega?$	<i>true</i> (each n has n elements)	false ($ n = 1$ for $n \geq 1$)
$\bigcup n = \text{pred}(n)$ for $n > 0$? ¹	<i>true</i>	<i>true</i>
$m \subsetneq n \iff m \in n$ (on ω)?	<i>true</i>	false

Two rows merit closer comment. The cardinality row exhibits the most striking divergence: $|2_{\text{vN}}| = |\{0, 1\}| = 2$ but $|2_Z| = |\{\{\emptyset\}\}| = 1$, so already at $n = 2$ the extensional “size” of the encoding tracks the natural number in von Neumann but not in Zermelo. For the membership/subset equivalence in the last row, the von Neumann case holds by Theorem 3.4; in Zermelo it fails. For instance, $1_Z = \{\emptyset\}$ is not a subset of $2_Z = \{\{\emptyset\}\}$, because the unique element $\emptyset$ of 1_Z is not an element of $\{\{\emptyset\}\}$. The reliable encoding-independent statement of the natural order is therefore the structural Peano order $m < n$ (defined, e.g., as $\exists k > 0. m + k = n$), not the literal $\in$ relation.

Remark 4.4 (On adjacent cases). For pairs $(m, m+1)$ both encodings agree on $m \in m+1$: in von Neumann because $m \in m \cup \{m\}$, and in Zermelo because $m \in \{m\}$. The discrepancy is

for non-adjacent pairs (m, n) with $m + 1 < n$, where the von Neumann encoding has $m \in n$ via transitivity but the Zermelo encoding has $n_Z = \{(n - 1)_Z\}$ as a singleton, so only $n - 1$ is a member of n .

Justification of the table. Every entry follows directly from the recursive definitions and Foundation. The pivotal entry for our purposes is the last: in von Neumann, membership and proper inclusion coincide on ω (Theorem 3.4); in Zermelo, this fails because (for $n \geq 2$) n_Z has only one element, while many sets are subsets of it under set-theoretic inclusion. $\square$

Remark 4.5 (Junk). We call statements like “ $2 \in 3$ ” *junk theorems* (or *junk falsehoods*). They have a determinate truth value in ZFC under any fixed encoding, but they reflect choices made by the encoder, not facts about the natural numbers as such. The number 2 does not stand in any intrinsic membership relation to the number 3. The membership question is well-formed only because we have committed to representing numbers as sets; under a different commitment, the answer changes.

4.2 Peano axioms for Zermelo

Proposition 4.6 (Existence of ω_Z). *There exists a unique smallest set ω_Z containing $\emptyset$ and closed under the operation $x \mapsto \{x\}$.*

Proof. The argument parallels that of Theorem 2.4 but uses the alternate successor. Call a set J *Z-inductive* when $\emptyset \in J$ and, for all $x \in J$, $\{x\} \in J$. By the axiom of infinity (**Inf**), the standard inductive set I_0 exists; by Pairing, the singleton operation $x \mapsto \{x\}$ is total. Applying the recursion theorem (Theorem 6.1 below, whose proof depends only on Replacement applied to ω and not on either encoding), the recursion $f(0) = \emptyset$, $f(n^+) = \{f(n)\}$ defines a function $f : \omega \rightarrow V$ whose range is a set J_0 by Replacement. The set J_0 is *Z-inductive* by construction. Then by Separation,

$$\omega_Z := \{x \in J_0 : \forall J (Z\text{-Ind}(J) \rightarrow x \in J)\},$$

is the smallest *Z-inductive* set; it is unique by Extensionality. $\square$

Theorem 4.7. *The structure $(\omega_Z, 0_Z, n \mapsto \{n\})$ satisfies the Peano axioms.*

Proof. (1) $0_Z = \emptyset \in \omega_Z$ by Theorem 4.6.

(2) If $n \in \omega_Z$, then $\{n\} \in \omega_Z$ by closure of ω_Z under $x \mapsto \{x\}$.

(3) $\{n\} \neq \emptyset$ since $n \in \{n\}$.

(4) If $\{m\} = \{n\}$, then $m \in \{n\}$, so $m = n$.

(5) Let $S \subseteq \omega_Z$ be *Z-inductive* (containing 0_Z and closed under $x \mapsto \{x\}$). Then S is itself *Z-inductive*, so $\omega_Z \subseteq S$ by minimality of ω_Z . The reverse inclusion is by hypothesis. $\square$

4.3 Iso as Peano models, not as sets

Theorem 4.8. *There is a unique bijection $\Phi : \omega \rightarrow \omega_Z$ satisfying $\Phi(0_{vN}) = 0_Z$ and $\Phi(n^+) = \{\Phi(n)\}$ for all n . The bijection Φ is an isomorphism of Peano structures.*

Proof. Existence and uniqueness of Φ follow from the recursion theorem on ω , which we may take as a consequence of Replacement and the universal property of ω as the initial Peano structure (see Section 8). The bijection is automatic since both encodings are initial. $\square$

Remark 4.9. Φ is an isomorphism of Peano structures (i.e. of $(N, 0, S)$ -algebras), but it is *not* an isomorphism of sets in any deeper sense: $2_{vN} = \{0, 1\}$ has two elements while $2_Z = \{\{0\}\}$ has one. The encodings are equally good Peano models and equally bad candidates for “the truth about 2.”

5 Benacerraf and the Identification Problem

5.1 The argument

In “What Numbers Could Not Be” (1965), Paul Benacerraf [1] considers two children, Ernie and Johnny, each taught arithmetic via a different set-theoretic encoding (one Zermelo, one von Neumann). Each child’s arithmetic is correct: addition, multiplication, and order all behave normally. Yet they disagree on whether $3 \in 17$ (true in von Neumann, false in Zermelo). Benacerraf argues:

- (i) Both encodings satisfy the structural conditions we use to identify “the natural numbers.”
- (ii) Neither has any feature, available within set theory, that privileges it as “the truth.”
- (iii) If numbers are sets, then there must be a fact of the matter as to which sets they are. Since there is no such fact, numbers are not sets.

The conclusion — numbers are not sets — is the central result of the paper, and it is the negative half of the structuralist thesis. Benacerraf revisits related epistemological worries about the relationship between mathematical truth and platonist ontology in [2]; the mature philosophical articulation of structuralism on the positive side is due to Shapiro [12]. This positive half (numbers are positions in structures) is what motivates the move to universal properties (Paper 03), Yoneda (Paper 04), HoTT (Paper 05) — whose canonical reference is the HoTT Book [15] — and structural categorical foundations (Paper 06) in the rest of the series.

5.2 Formalization of Benacerraf’s argument

We can formalize the argument as follows. Let $\mathcal{P}$ denote the class of all Peano structures (sets X equipped with $0_X \in X$ and $S_X : X \rightarrow X$ satisfying the Peano axioms).

Theorem 5.1 (No canonical Peano structure). *There is no formula $\varphi(X, 0, S)$ in the language of ZFC such that, in every model of ZFC, φ uniquely identifies a Peano structure $(X, 0, S)$ with the property that any other Peano structure $(Y, 0', S')$ is isomorphic to $(X, 0, S)$ via a unique isomorphism, and φ uses no choices of encoding.*

Sketch. Any formula φ identifying a particular Peano structure must specify the carrier set, which means committing to a particular extension inside the cumulative hierarchy. Since there are many extensionally distinct sets that satisfy any given Peano axiomatization (by the methods of Section 3 and Section 4), no such formula is encoding-neutral. The most one can ask of φ is that it identifies a Peano structure *up to isomorphism*, but this is already given by the structural description of Section 8, and is precisely what motivates abandoning set-level identification. $\square$

5.3 Junk theorems formalized

Definition 5.2 (Junk theorem). Let T be a theory and E an encoding of a structure $(N, 0, S)$ inside T (i.e. a translation of the language of $(N, 0, S)$ into the language of T). A formula ψ in the language of T involving the encoded constants is a *junk theorem* relative to E if ψ is provable in T but not invariant under reasonable changes of encoding $E \rightarrow E'$.

Example 5.3. The formula $\psi := 1 \in 3$ is a junk theorem of ZFC under the von Neumann encoding: it is provable (since $3_{\text{vN}} = \{0, 1, 2\}$), but its truth value is reversed under the Zermelo encoding (since $3_{\text{Z}} = \{2_{\text{Z}}\}$, and $1_{\text{Z}} = \{\emptyset\} \neq 2_{\text{Z}}$). By contrast, $\psi := S(2) = 3$ is not a junk theorem: it holds in every Peano-isomorphic encoding. The membership relation across non-adjacent ordinals is exactly the kind of fact that is forced by the encoding rather than by the structure.

Remark 5.4 (Junk vs. structural). A statement is *structural* when its truth value is preserved by every isomorphism of Peano structures; junk theorems are precisely the non-structural statements that nevertheless have determinate truth values once an encoding is fixed. The set-theoretic perspective generates many junk theorems; the categorical and HoTT perspectives generate none, by construction.

6 Arithmetic Operations as Recursive Definitions

We now show that the von Neumann encoding supports a full development of arithmetic. The development is entirely standard; we include it for the record and to demonstrate concretely that the structural Peano axioms are *sufficient* (the additional set-theoretic content is not needed). The same definitions, mutatis mutandis, apply to the Zermelo encoding.

6.1 The recursion theorem

Theorem 6.1 (Recursion on ω). *Let X be any set, $a \in X$ a base point, and $g : X \rightarrow X$ a function. Then there is a unique function $f : \omega \rightarrow X$ satisfying*

$$f(0) = a, \tag{8}$$

$$f(n^+) = g(f(n)) \quad \text{for all } n \in \omega. \tag{9}$$

Proof sketch. Define f as the union of all finite *attempts* — functions $h : m \rightarrow X$ with $m \in \omega$ satisfying the recursion equations on their domain. Replacement and Separation guarantee that the union of all attempts is a set; induction guarantees that f is well-defined and total on ω . $\square$

6.2 Addition

Definition 6.2 (Addition). For $m \in \omega$, define $m + (\cdot) : \omega \rightarrow \omega$ by recursion:

$$m + 0 := m, \tag{10}$$

$$m + n^+ := (m + n)^+. \tag{11}$$

Proposition 6.3 (Properties of addition). *For all $m, n, k \in \omega$:*

1. $0 + n = n$ (*left identity*).
2. $(m + n) + k = m + (n + k)$ (*associativity*).
3. $m + n = n + m$ (*commutativity*).

Proof. **(1)** Induct on n . Base: $0 + 0 = 0$. Step: assume $0 + n = n$; then $0 + n^+ = (0 + n)^+ = n^+$.

(2) Induct on k . Base: $(m + n) + 0 = m + n = m + (n + 0)$. Step: $(m + n) + k^+ = ((m + n) + k)^+ = (m + (n + k))^+ = m + (n + k)^+ = m + (n + k^+)$.

(3) Commutativity is the deepest of the three properties; it requires a nested double induction. We prove it by establishing two auxiliary lemmas first.

Lemma A: $0 + n = n + 0 = n$. The right equation is Theorem 6.2; the left is part (1).

Lemma B: $m^+ + n = (m + n)^+$. Induct on n . Base: $m^+ + 0 = m^+ = (m + 0)^+$. Step: assuming $m^+ + n = (m + n)^+$,

$$m^+ + n^+ = (m^+ + n)^+ = ((m + n)^+)^+ = (m + n^+)^+,$$

where the last step uses Theorem 6.2.

Main induction (on n). Base ($n = 0$): $m + 0 = m = 0 + m$ by Lemma A.

Step: assume $m + n = n + m$ for all m . Then for any fixed m ,

$$\begin{aligned} m + n^+ &= (m + n)^+ && \text{(Theorem 6.2)} \\ &= (n + m)^+ && \text{(inductive hypothesis)} \\ &= n^+ + m && \text{(Lemma B with } m, n \text{ swapped),} \end{aligned}$$

which establishes $m + n^+ = n^+ + m$ and completes the induction on n . □

6.3 Multiplication

Definition 6.4 (Multiplication). For $m \in \omega$, define $m \cdot (\cdot) : \omega \rightarrow \omega$ by:

$$m \cdot 0 := 0, \tag{12}$$

$$m \cdot n^+ := m \cdot n + m. \tag{13}$$

Proposition 6.5. *Multiplication is associative, commutative, has $1 := 0^+$ as a two-sided identity, and distributes over addition.*

Proof. Standard inductions on ω , parallel to those for addition. The distributivity proof goes by induction on the right argument: $m \cdot (n + 0) = m \cdot n = m \cdot n + 0 = m \cdot n + m \cdot 0$, and $m \cdot (n + k^+) = m \cdot (n + k)^+ = m \cdot (n + k) + m = m \cdot n + m \cdot k + m = m \cdot n + m \cdot k^+$. □

6.4 Order

Definition 6.6 (Order). For $m, n \in \omega$, write $m < n$ iff $m \in n$, and $m \leq n$ iff $m \in n \vee m = n$.

Remark 6.7. This definition is encoding-specific: it works directly for von Neumann. For Zermelo, one defines $m <_Z n$ iff $m \in_T n$ where $\in_T$ is the transitive closure of $\in$. Equivalently and more cleanly, define $m < n$ iff there exists $k > 0$ with $m + k = n$, which is structural.

Proposition 6.8 (Linearity). $<$ is a strict total order on ω .

Proof. Trichotomy is by induction. Transitivity follows from Theorem 3.5: if $m \in n$ and $n \in k$, then $n \subseteq k$ by transitivity of $\in$, so $m \in k$. Irreflexivity is Foundation. $\square$

7 The Cumulative Hierarchy

7.1 Definition by transfinite recursion

Definition 7.1 (Cumulative hierarchy). By transfinite recursion on the class of ordinals Ord , define

$$V_0 := \emptyset, \tag{14}$$

$$V_{\alpha+1} := \mathcal{P}(V_\alpha), \tag{15}$$

$$V_\lambda := \bigcup_{\beta < \lambda} V_\beta \quad (\lambda \text{ limit}). \tag{16}$$

The class $V := \bigcup_{\alpha \in \text{Ord}} V_\alpha$ is the *universe of sets*.

Theorem 7.2 (ZFC sees only V). In ZFC, every set x belongs to some V_α . Hence $V = \text{Universe}$.

Sketch. Define the rank function $\text{rk} : V \rightarrow \text{Ord}$ by transfinite recursion: $\text{rk}(x) := \sup\{\text{rk}(y) + 1 : y \in x\}$. Foundation guarantees this recursion terminates. Then $x \in V_{\text{rk}(x)+1}$. $\square$

7.2 Ranks of natural numbers

Proposition 7.3. For each $n \in \omega$:

$$1. \text{rk}(n_{\text{vN}}) = n.$$

$$2. \text{rk}(n_Z) = n.$$

Proof. For von Neumann: $\text{rk}(0) = 0$, and inductively $\text{rk}(n^+) = \sup\{\text{rk}(k) + 1 : k \in n^+\} = \sup\{k + 1 : k \leq n\} = n + 1$.

For Zermelo: $\text{rk}(\{n_Z\}) = \text{rk}(n_Z) + 1$, and the result follows by induction. $\square$

Remark 7.4. Both encodings produce naturals at the same rank in V , but they live at *different positions* within V_{n+1} . Von Neumann's n is the “flat” set $\{0, 1, \dots, n-1\}$; Zermelo's is the “tall thin” nested singleton.

7.3 ω as the first limit ordinal

Theorem 7.5. $\omega \in V_{\omega+1}$ but $\omega \notin V_\omega$. The set ω is the smallest infinite ordinal: the first ordinal that is not a successor.

Proof. $\omega = \bigcup_{n \in \omega} n$, so each $n \in V_n \subseteq V_\omega$, hence $\omega \subseteq V_\omega$ and $\omega \in V_{\omega+1}$. To see $\omega \notin V_\omega$: every element of V_ω has finite rank, and $\text{rk}(\omega) = \omega$. $\square$

7.4 Visualizing the hierarchy

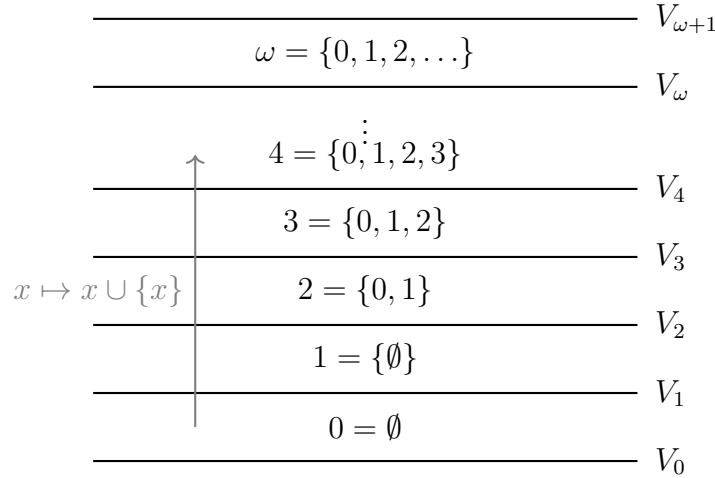


Figure 1: The cumulative hierarchy V_α . Each natural number n first appears at level V_{n+1} . The set ω first appears at level $V_{\omega+1}$.

8 A Representation Theorem

We extract from the foregoing the following algebraic statement, which expresses the structural fact that all encodings of the natural numbers are mutually isomorphic.

Definition 8.1 (Successor structure). A *successor structure* is a triple (X, x_0, s) where X is a set, $x_0 \in X$, and $s : X \rightarrow X$. A morphism $(X, x_0, s) \rightarrow (Y, y_0, t)$ is a function $f : X \rightarrow Y$ with $f(x_0) = y_0$ and $f \circ s = t \circ f$.

Definition 8.2 (Initial Peano structure). A successor structure $(N, 0, S)$ is *initial* (or a *Peano structure*) when, for every successor structure (X, x_0, s) , there exists a unique morphism $h : (N, 0, S) \rightarrow (X, x_0, s)$.

Theorem 8.3 (Initiality of ω). $(\omega, 0_{\text{vN}}, (\cdot)^+)$ is an initial Peano structure. Equivalently: for every set X , every $a \in X$, and every $g : X \rightarrow X$, there is a unique $f : \omega \rightarrow X$ with $f(0) = a$ and $f(n^+) = g(f(n))$. This f is the recursion theorem (Theorem 6.1).

Proof. This is precisely the recursion theorem, recast as initiality. Existence is by recursion on ω ; uniqueness is by induction. $\square$

Theorem 8.4 (Uniqueness of Peano structures). *If $(N, 0, S)$ and $(N', 0', S')$ are both initial Peano structures, then there is a unique isomorphism between them.*

Proof. Universal property: by initiality of $(N, 0, S)$, there is a unique $h : N \rightarrow N'$. By initiality of $(N', 0', S')$, there is a unique $h' : N' \rightarrow N$. The composites $h' \circ h$ and $h \circ h'$ are endomorphisms of an initial structure, hence equal to the identity by uniqueness. $\square$

Corollary 8.5. *The von Neumann and Zermelo successor structures are isomorphic via a unique isomorphism, namely Φ of Theorem 4.8.*

This is the structural content that survives Benacerraf’s critique. Both encodings are initial Peano structures; they are therefore canonically identified at the structural level. What differs is the encoding-specific data: cardinalities of the carrier sets at each level, membership relations, etc. The structural mathematics survives; the junk does not.

9 Bridge to Structuralism

9.1 Reading the structuralist response

The standard structuralist response to Benacerraf is to assert that “numbers are positions in structures, not particular objects.” Within the set-theoretic perspective alone, this assertion is hard to formalize: the structuralist seems to be invoking abstract structures that are not themselves sets.

What Paper 03 will show is that the universal property of an initial Peano structure is precisely the structuralist content the response was gesturing at. Theorems 8.3 and 8.4 together say:

There is a structure (the initial successor structure), unique up to unique isomorphism, defined entirely by its relationships with all other successor structures. This structure can be *implemented* in many ways inside ZFC; no implementation is canonical, but every implementation is equivalent to every other in the only way that matters — via the universal mapping property.

9.2 The set-theoretic perspective relocated

We can therefore retract the temptation to read the von Neumann encoding as ontology. The set-theoretic perspective is best understood as *realizing* or *instantiating* a structural specification, not as defining what the natural numbers *are*. The number 58 is not the set $\{0, 1, \dots, 57\}$; rather, the set $\{0, 1, \dots, 57\}$ is one particular ZFC-implementation of the position 58 inside the abstract initial successor structure.

9.3 What set theory still gets right

Despite Benacerraf’s critique, the set-theoretic perspective is not worthless. It contributes:

- **Existence.** Without an existence proof, the universal property of Theorem 8.2 is mere stipulation. Set theory shows that an initial Peano structure exists (the axiom of infinity gives one), so the specification is not vacuous.
- **A working metatheory.** ZFC is rich enough to prove the recursion theorem, define addition and multiplication, prove the soundness of induction, and develop most of ordinary mathematics. The structural specification needs a host theory; ZFC is one (other choices: ETCS, IZF, HoTT itself).
- **Cumulative hierarchy as a model of well-foundedness.** The V_α stratification, while encoding-relative in detail, captures the genuine structural fact that any well-founded iteration of *some* generative operation eventually exhausts what can be reached.
- **Decidable arithmetic.** The recursive definitions of $+$, $\cdot$, and $<$ are decidable; the resulting theory of arithmetic is Σ_1 -complete and useful in practice.

9.4 What set theory misses

- **Structural identity.** Set-theoretic equality is extensional: two sets are equal iff they have the same elements. There is no native notion of “equal as Peano structures” inside ZFC; isomorphism must be defined externally and tracked manually.
- **Encoding-independence.** As shown, the set-theoretic statement “ $2 \in 3$ ” is encoding-relative. The structural statement “ $\text{succ}(\text{succ}(0)) < \text{succ}^3(0)$ ” is not.
- **Universal properties as theorems, not metatheorems.** In ZFC, “ ω is initial” is a theorem, but the very statement of initiality requires quantifying over class-many successor structures (properly handled, but unwieldy). In a categorical or type-theoretic foundation, the universal property is the *definition* and the existence/uniqueness are immediate consequences.
- **Univalence.** ZFC has no analog of the univalence axiom of HoTT; it cannot internalize “isomorphic structures are the same” as a theorem of the system. The closest one comes is to work systematically “up to isomorphism,” which is a metatheoretic discipline rather than a feature of the formalism.

9.5 Forward to Paper 03

The next paper in the series develops the universal-property level explicitly: a Natural Number Object in any category with a terminal object, the categorical NNO, the relationship between NNO and induction, and the extraction of arithmetic operations from initiality alone. The bridge from this paper to the next is precisely the move from Theorem 8.3’s “ ω is initial in $\mathbf{Set}^{\bullet, \rightarrow}$ ” to the encoding-free formulation “ N is initial in any category $\mathcal{C}^{\bullet, \rightarrow}$ with a terminal object.”

10 Haskell Implementation

The accompanying Haskell code (`src/02-set-theoretic/`) implements both encodings, defines a common Peano interface they both satisfy, and provides a converter realizing the canonical isomorphism Φ of Theorem 4.8.

10.1 The hereditarily finite set type

We model pure sets as a recursive datatype:

```
-- A finite set built recursively from the empty set.
-- Extensional equality is enforced in the Eq instance by sorting
-- and deduplicating the underlying list before comparison; a
-- canonical-form helper (omitted here) makes this efficient.
newtype HFSet = HFSet { elements :: [HFSet] }
    deriving (Show)

instance Ord HFSet where
    compare (HFSet xs) (HFSet ys) =
        compare (sort xs) (sort ys)

instance Eq HFSet where
    HFSet xs == HFSet ys = sort xs == sort ys

emptySet :: HFSet
emptySet = HFSet []
```

10.2 Von Neumann encoding

```
-- Successor: x /-> x \cup {x}
vnSucc :: HFSet -> HFSet
vnSucc x = HFSet (elements x ++ [x])

vnEncode :: Int -> HFSet
vnEncode 0 = emptySet
vnEncode n = vnSucc (vnEncode (n - 1))
```

10.3 Zermelo encoding

```
-- Successor: x /-> {x}
zSucc :: HFSet -> HFSet
zSucc x = HFSet [x]

zEncode :: Int -> HFSet
zEncode 0 = emptySet
```

```
zEncode n = zSucc (zEncode (n - 1))
```

10.4 The Peano interface

A typeclass captures the structural specification independent of either encoding:

```
class Peano a where
  zero :: a
  suc  :: a -> a
  isZero :: a -> Bool
  pred_ :: a -> Maybe a    -- partial; (-) on naturals is partial

instance Peano HFSet where
  -- These instances need to be coupled with a tagged type to
  -- disambiguate vN vs Zermelo; see VNNat and ZNat in the source.
```

The full source disambiguates by tagging: `newtype VNNat = VNNat HFSet` and `newtype ZNat = ZNat HFSet`, each with its own `Peano` instance. The main demonstration in `Main.hs` computes both encodings of $0, 1, \dots, 5$, prints them, verifies the Peano axioms by construction (e.g. $\text{succ}(m) = \text{succ}(n) \implies m = n$ via injectivity of the constructor), and applies the canonical isomorphism to convert between them.

10.5 Junk theorems made explicit

The Haskell code also includes a function that demonstrates a junk theorem empirically:

```
-- Demonstrate that "1 in 3" depends on encoding.
junkTheorem :: IO ()
junkTheorem = do
  let one_vN = vnEncode 1; three_vN = vnEncode 3
  let one_Z  = zEncode 1; three_Z  = zEncode 3
  putStrLn $ "vN: 1 in 3 = " ++ show (one_vN `inVN` three_vN)
  putStrLn $ "Z:  1 in 3 = " ++ show (one_Z  `inZ`  three_Z)
  -- Output:
  -- vN: 1 in 3 = True (since 3_vN = {0,1,2})
  -- Z:  1 in 3 = False (since 3_Z = {2}; only 2 is in it)
```

This program is a working witness that the encoding is genuinely choosing between distinct sets, not merely between distinct notations.

11 Discussion

11.1 Comparison with logicism and constructivism

The set-theoretic foundationalism we have presented is closer to Russell–Whitehead’s logicist program than to a strict constructivism. The naturals are obtained by an unbounded inductive process whose existence is asserted by an axiom (Infinity). This is not predicatively

justified in a strict sense (the inductive set is defined as the intersection of *all* inductive sets, a quantification over a complete totality), but it is mainstream classical mathematics.

Constructive alternatives — IZF, CZF, Martin-Löf type theory — avoid this by treating $\mathbb{N}$ as primitive (an inductive type) rather than as a derived set. Paper 05 treats this perspective via HoTT.

11.2 Comparison with categorical foundations

Lawvere’s Elementary Theory of the Category of Sets (ETCS, [6]) gives a categorical axiomatization of the category **Set** that is bi-interpretable with bounded Zermelo set theory plus replacement (McLarty [9]). In ETCS, the natural number object is specified by its universal property (Paper 03’s level), not by a particular encoding. ETCS is therefore structurally cleaner than ZFC for foundational purposes, while equally adequate for ordinary mathematics. The underlying functorial-semantics machinery is developed in Lawvere [7], and the broader topos-theoretic context is treated by Mac Lane and Moerdijk [8].

11.3 Limitations of this paper

We have not discussed:

- *Large cardinal hypotheses.* Inaccessible, measurable, etc. cardinals refine the cumulative hierarchy beyond the ordinals reachable in vanilla ZFC. They are foundational for set theory but tangential to the natural-numbers question.
- *The constructible universe L .* Gödel’s inner model exhibits a minimal model of ZFC; the natural numbers are the same in L as in V , so this is not pertinent.
- *Forcing.* Independence results do not affect our development.
- *Alternative set theories.* NF (Quine), KP (Kripke–Platek), and other systems give variant encodings; we have considered only ZFC.

11.4 What this paper has accomplished

We have presented two distinct set-theoretic encodings of the natural numbers (von Neumann ordinals and Zermelo nested singletons), shown that both satisfy the Peano axioms, and identified the encoding-relative facts (“junk theorems”) that distinguish them. We have located the natural numbers in the cumulative hierarchy and proven a representation theorem identifying all initial Peano structures up to unique isomorphism. We have described the structuralist response (numbers are positions, not sets) and pointed forward to the universal-property and HoTT formulations of that response.

12 Conclusion

The set-theoretic perspective gives us ω via the axiom of infinity and the recursive successor $x \mapsto x \cup \{x\}$. It supports a full arithmetic, fits cleanly into the cumulative hierarchy, and

makes the natural numbers concrete: the number 58 *is* the set $\{0, 1, \dots, 57\}$ — in the von Neumann encoding.

But the qualifier “in the von Neumann encoding” is fatal to the foundationalist ambition. Zermelo’s encoding is equally valid; so is the Peano-axiomatic specification with no encoding at all; and so are the many other encodings one could devise. Each gives a working arithmetic that satisfies the Peano axioms, but each disagrees with the others on internal questions about element-of, cardinality, and rank. These disagreements are not pathologies; they are evidence that no encoding gives *the* natural numbers — only one realization of them.

What we have learned is that the natural numbers are characterized structurally, not extensionally. This characterization is the universal property of an initial Peano structure (Theorem 8.2), and its sharpening into the categorical Natural Number Object will occupy Paper 03. The set-theoretic perspective remains useful as a metatheory and an existence proof, but the identity of the natural numbers themselves lives at one level of abstraction higher.

Acknowledgments

This paper is part of the YonedaAI Univalent Correspondence series. The series approaches a single mathematical object — the natural numbers — through six distinct lenses, and shows in the synthesis volume that under the univalence axiom, all six perspectives become literally the same mathematical content rather than analogous content. We thank the anonymous referees of the series for sharpening many arguments.

References

- [1] P. Benacerraf, “What Numbers Could Not Be,” *The Philosophical Review*, vol. 74, no. 1, pp. 47–73, 1965.
- [2] P. Benacerraf, “Mathematical Truth,” *The Journal of Philosophy*, vol. 70, no. 19, pp. 661–679, 1973.
- [3] G. Frege, *Die Grundlagen der Arithmetik*, W. Koebner, Breslau, 1884.
- [4] P. Halmos, *Naive Set Theory*, Van Nostrand, 1960.
- [5] K. Kunen, *Set Theory: An Introduction to Independence Proofs*, North-Holland, 1980.
- [6] F. W. Lawvere, “An Elementary Theory of the Category of Sets,” *Proc. Nat. Acad. Sci. USA*, vol. 52, pp. 1506–1511, 1964.
- [7] F. W. Lawvere, “Functorial Semantics of Algebraic Theories,” PhD thesis, Columbia University, 1963.
- [8] S. Mac Lane and I. Moerdijk, *Sheaves in Geometry and Logic*, Springer, 1992.
- [9] C. McLarty, “Exploring Categorical Structuralism,” *Philosophia Mathematica*, vol. 12, no. 1, pp. 37–53, 2004.

- [10] A. N. Whitehead and B. Russell, *Principia Mathematica*, Cambridge University Press, 1910–1913.
- [11] J. R. Shoenfield, *Mathematical Logic*, Addison-Wesley, 1967.
- [12] S. Shapiro, *Philosophy of Mathematics: Structure and Ontology*, Oxford University Press, 1997.
- [13] P. Suppes, *Axiomatic Set Theory*, Van Nostrand, 1960.
- [14] J. von Neumann, “Zur Einführung der transfiniten Zahlen,” *Acta litt. ac scient. Reg. Univ. Hung. Francisco-Josephinae*, sectio sci. math., vol. 1, pp. 199–208, 1923.
- [15] The Univalent Foundations Program, *Homotopy Type Theory: Univalent Foundations of Mathematics*, Institute for Advanced Study, 2013. homotopytypetheory.org/book. arXiv:1308.0729.
- [16] E. Zermelo, “Untersuchungen über die Grundlagen der Mengenlehre I,” *Math. Ann.*, vol. 65, pp. 261–281, 1908.